

Nutanix AHV + Cloudera

VM Placement & Maintenance Guidelines

Version 1.0 | May 2026 | For Review

1. Purpose & Scope

This document provides operational best-practice guidelines for the deployment and ongoing maintenance of Cloudera-based virtual machines (VMs) on a Nutanix AHV platform. It covers two cluster types:

- Base Cluster: Cloudera Data Platform (CDP) master and worker VMs running core services (HDFS, YARN, ZooKeeper, Cloudera Manager, etc.)
- DS Cluster (Data Services): Cloudera Data Services running on Embedded Container Service (ECS), including control-plane and worker node VMs.

The primary goal is to ensure that planned maintenance of a physical Nutanix host (AHV node) can be performed without interrupting Cloudera workloads, Kubernetes quorum, or data availability. These guidelines focus on VM affinity/anti-affinity policy configuration and cluster capacity sizing.

Note: The deployment and functional testing previously conducted covered the Cloudera stack installation and validation of core services. It did not specifically cover maintenance-mode scenarios, VM live migration, or host-failure simulations. The recommendations in this document are based on Nutanix AHV and Cloudera architectural best practices.

2. Key Concepts

2.1 VM-VM Anti-Affinity Policy

A VM-VM Anti-Affinity policy is a placement rule configured in Nutanix Prism Central that instructs the AHV scheduler to ensure a defined group of VMs are always placed on different physical hosts. This is the primary mechanism to protect high-availability (HA) service pairs or quorum-based clusters from a single point of failure at the physical host level.

For Cloudera, this is critical for roles deployed in pairs or odd-numbered quorum groups, such as:

- HDFS NameNode (Active + Standby): both nodes must never reside on the same AHV host.
- ZooKeeper ensemble (3 or 5 nodes): losing a majority of ZooKeeper nodes (e.g., 2 of 3) causes complete cluster loss; anti-affinity across hosts prevents this.
- YARN ResourceManager (Active + Standby): ensures scheduling continuity.
- ECS Control-Plane masters (3 nodes): etcd quorum requires at least 2 of 3 masters to be reachable at all times.

Navigation in Prism Central:

| Prism Central -> Policies -> VM Policies -> VM-VM Anti-Affinity Policies -> Create Policy

2.2 N+1 Host Survivability (Capacity Headroom)

N+1 host survivability is an industry-standard cluster sizing model where N is the number of active physical hosts required to run all workloads, and the cluster is provisioned with one additional host worth of spare CPU and RAM capacity at all times.

In practice, this means:

- When one physical host is placed into planned maintenance mode (or fails unexpectedly), the Nutanix cluster has sufficient free resources across the remaining hosts to live-migrate all VMs that were running on the offline host.
- No VM needs to be powered off, and no Cloudera service is interrupted during the evacuation.
- The spare capacity headroom must account for the largest single host in the cluster (the host with the most VMs assigned to it at peak).

| *Best Practice: Do not consistently run the Nutanix cluster at above 70–75% aggregate CPU or RAM utilization. Reserve the remaining 25–30% as headroom for VM migration and burst workloads, in addition to maintaining formal N+1 sizing.*

2.3 AHV Live Migration

Nutanix AHV supports non-disruptive live migration of powered-on VMs between physical hosts. When a host is placed into maintenance mode via Prism Element or Prism Central, AHV will automatically attempt to live-migrate all running VMs to other hosts in the cluster, provided sufficient capacity exists. This process is transparent to the guest OS and running applications.

3. VM Affinity/Anti-Affinity Policy Recommendations

The table below summarises the recommended VM placement policies for all key VM types in both the Base and DS clusters:

VM Category	Policy Type	Rationale
HDFS NameNode (Active + Standby)	VM-VM Anti-Affinity	Ensures HA; prevents both nodes on same AHV host
ZooKeeper nodes (3 or 5)	VM-VM Anti-Affinity	Maintains quorum if one host goes offline
YARN ResourceManager (Active + Standby)	VM-VM Anti-Affinity	Maintains cluster scheduling continuity
ECS Control-Plane masters (3 nodes)	VM-VM Anti-Affinity	Prevents loss of etcd/control-plane quorum
Cloudera Manager Server	Preferred Anti-Affinity or standalone	Reduces single-host risk for management plane
Base Cluster Worker VMs	No strict affinity (balanced placement)	Maximum migration flexibility during maintenance
DS (ECS) Worker Nodes	No strict affinity (balanced placement)	Allows AHV to rebalance across remaining hosts

Worker VMs (Base cluster DataNodes and DS cluster worker nodes) do not require strict affinity rules. Allowing balanced and flexible placement maximises AHV's ability to rebalance load and perform live migrations during maintenance windows.

4. Capacity Sizing Recommendations

The following table summarises the capacity headroom recommendations to enable smooth maintenance operations:

Parameter	Recommended Approach	Notes
CPU Headroom	N+1 spare capacity (1 full host equivalent)	Sufficient to absorb all VMs from one host
RAM Headroom	N+1 spare capacity (1 full host equivalent)	Allows live migration without memory overcommit
Storage / CVM	RF2 or RF3 depending on risk tolerance	Recommended is RF2 for Cloudera environments

Parameter	Recommended Approach	Notes
Network	Dedicate separate vLAN / vNIC for Cloudera data traffic	Reduces contention during VM migration events

4.1 Estimating Required Headroom

To calculate the required spare capacity for N+1 survivability, identify the host with the highest aggregate vCPU and RAM allocation across its assigned VMs. The cluster must have at least that amount of free CPU and RAM distributed across the remaining hosts at all times.

Example (illustrative):

- Cluster: 4 physical hosts, each with 256 GB RAM and 48 physical cores.
- One host runs: 2x NameNode (64 GB RAM, 16 vCPU) + ZooKeeper (16 GB, 4 vCPU) + 3x Workers (48 GB, 12 vCPU each) = total ~224 GB RAM, ~56 vCPU.
- The remaining 3 hosts must collectively have at least 224 GB RAM and 56 vCPU free to absorb those VMs without disruption.

5. Recommended Maintenance Mode Procedure

When performing planned maintenance on a physical Nutanix host, the following sequence is recommended:

- Pre-check: Verify cluster free CPU and RAM are sufficient for N+1 evacuation via Prism Central dashboards.
- Pre-check: Confirm all anti-affinity policies are healthy (no policy violations currently exist).
- Enable Maintenance Mode: In Prism Element, select the target host and enable Maintenance Mode. AHV will initiate live migration of all VMs.
- Monitor: Watch VM migration progress in Prism and monitor Cloudera Manager for any service alerts. Anti-affinity policies will be honoured during migration placement.
- Maintenance: Perform the required physical host maintenance.
- Exit Maintenance Mode: Re-enable the host in Prism. AHV will automatically rebalance VMs if the cluster is configured for DRS-equivalent rebalancing.
- Post-check: Validate all Cloudera services are green in Cloudera Manager and all Kubernetes nodes are Ready in the DS cluster.

Important: If anti-affinity policies cannot be satisfied during VM migration (e.g., insufficient hosts to spread all anti-affined VMs), AHV will place a VM in violation of the policy rather than leave it unpowered. Monitor for policy violations post-maintenance and review sizing if violations occur regularly.

6. VM Disk Type Recommendations

On Nutanix AHV, there are two distinct ways to present storage to a virtual machine. Understanding the difference between these types — and the concept of dependent vs. independent disks — is essential for correct snapshot behaviour, storage performance, and operational flexibility across Cloudera workloads.

6.1 Dependent vs. Independent Disks: The Core Concept

In VMware terminology, a dependent disk is included in VM-level snapshots, whereas an independent (persistent) disk is excluded. Nutanix AHV maps to the same concept as follows:

- Standard vDisk (dependent equivalent): included in VM-level Nutanix snapshots and Protection Domain policies. Used for OS disks and service-configuration disks where consistent point-in-time recovery of the full VM is required.
- Volume Group (independent/persistent equivalent): excluded from VM-level snapshots by default. VGs have their own independent snapshot and Protection Domain lifecycle, separate from the VM. This is the recommended approach for data disks that should not be tied to VM-level snapshot schedules — particularly HDFS DataNode data volumes.

Key Rule: Attach DataNode data volumes as Volume Groups (VGs). This is the Nutanix AHV equivalent of VMware independent persistent disks — they are excluded from VM-level snapshots, preventing snapshot overhead from impacting HDFS data I/O and avoiding bloated, inconsistent snapshots of large data volumes.

6.2 Standard vDisk

A standard vDisk is a virtual disk attached directly to a VM through the AHV virtual SCSI or VirtIO controller, stored as an object in the Nutanix Distributed Storage Fabric (DSF). It is the default disk type for OS and application disks.

Key characteristics:

- Managed through Prism Element or Prism Central; supports resize, clone, and snapshot at the VM level.
- Included in VM-level snapshots and Protection Domain policies — suitable for OS and config disks.
- OS vDisks support flexible partition management inside the guest OS, allowing resizing without VM downtime.
- Travels with the VM transparently during AHV live migration.

6.3 Volume Group (Nutanix Volumes)

A Volume Group (VG) is a named container of one or more vDisks, presented to a VM as independent block devices via direct AHV SCSI attachment. On AHV, no iSCSI initiator configuration is required inside the guest — the VG disks appear directly as SCSI devices (e.g., /dev/sdb, /dev/sdc).

Key characteristics:

- Each vDisk in the VG appears as a separate block device inside the guest OS — HDFS manages its own data directories directly, one per disk.
- Excluded from VM-level snapshots by default — VG lifecycle (snapshots, clones, Protection Domains) is managed independently from the VM.
- Supports Volume Group Load Balancing (VGLB) on AHV, distributing vDisk I/O ownership across multiple CVMs for high-throughput workloads.
- Use multiple smaller disks per VG rather than one large disk to distribute I/O across Nutanix CVMs (aim for one disk per Nutanix node as a minimum baseline; 4–8 disks per DataNode VM for production HDFS deployments).

6.4 Recommended Disk Layout per VM Role

The table below summarises the recommended disk type and storage tier for each VM role:

VM Role	Disk Type	Notes
Cloudera Manager Server	Standard vDisk	1x OS + 1x data vDisk (DB, logs, parcels)
HDFS NameNode (Active + Standby)	Standard vDisk	1x OS + 1x metadata/edits vDisk
HDFS JournalNodes	Standard vDisk	1x OS + 1x journal edits vDisk
ZooKeeper nodes	Standard vDisk	1x OS + 1x vDisk for dataDir/dataLogDir
YARN ResourceManager	Standard vDisk	1x OS vDisk (stateless role)
ECS / K8s Control-Plane masters	Standard vDisk	1x OS + 1x dedicated etcd vDisk
Base Cluster Worker VMs (DataNodes)	Standard vDisk (OS) + Volume Group (data)	1x OS vDisk + 1 VG with 4–8 data vDisks
DS / ECS Worker Nodes	Standard vDisk (OS) + Standard vDisk (Longhorn)	1x OS vDisk + 1x vDisk
NiFi nodes	Standard vDisk (OS) + Volume Group (repos)	1x OS vDisk + separate VG disk per repo

7. Summary of Key Recommendations

- Configure VM-VM Anti-Affinity policies in Prism Central for all Cloudera HA master pairs and quorum nodes (NameNode, ZooKeeper, ResourceManager, ECS/K8s masters).
- Size the Nutanix cluster with N+1 host survivability: the cluster must have at least one full host's worth of free CPU and RAM at all times.
- Avoid strict host affinity rules for worker VMs (DataNodes, ECS/K8s workers) to maximise live migration flexibility.
- Keep aggregate cluster utilisation below 70–75% to maintain effective headroom for migrations and burst activity.
- Follow the structured maintenance mode procedure (pre-check, enable maintenance, monitor, exit, post-check) for all planned physical host operations.
- Use standard vDisks for all OS and control-plane data disks; use Volume Groups (independent from VM snapshots) for HDFS DataNode data directories and ECS/K8s worker data. For ZooKeeper, ensure dataLogDir is on an SSD-backed vDisk. For NiFi, provision each repository on a separate SSD-backed vDisk with the Content Repository on multiple disks.

8. References

The following Nutanix and Cloudera official documentation was referenced in preparing these guidelines:

- Nutanix AHV Administration Guide v11.0: VM-VM Anti-Affinity Policies — portal.nutanix.com/docs/AHV-Admin-Guide-v11_0:ahv-anti-affinity-policies-c.html
- Nutanix AHV Administration Guide v11.0: VM-VM Anti-Affinity Policy Configuration (Prism Central) — portal.nutanix.com/docs/AHV-Admin-Guide-v11_0:ahv-vm-anti-affinity-t.html
- Nutanix AHV Administration Guide v10.3: Node Maintenance Mode — portal.nutanix.com/docs/AHV-Admin-Guide-v10_3:wc-node-maintenance-ahv-wc.html
- Nutanix Support KB: How to Place a CVM and Host in Maintenance Mode — portal.nutanix.com/page/documents/kbs/details/?targetId=kA00e000000LIQoCAO
- Cloudera Runtime 7.3.1: Enabling HDFS High Availability and Automatic Failover — docs.cloudera.com/runtime/7.3.1/fault-tolerance/topics/cr-enabling-high-availability-and-automatic-failover.html
- Cloudera ECS 1.5.5: Enabling ECS Server HA Post Installation — docs.cloudera.com/cdp-private-cloud-data-services/1.5.5/installation-ecs/topics/cdppvc-install-ecsha-0.html

For more documentations, refer to: portal.nutanix.com and docs.cloudera.com